

A Ring Lemma for Band-Limited Vorticity Direction and a Conditional Spectral Non-Dispersal Criterion

on the Three-Dimensional Torus

Jonathan Robert Simons
Savannah, Georgia, USA
simonsmedicalinnovations@gmail.com

Corrected preprint — August 2026

Disclaimer. The main regularity statement is *conditional* on Spectral Non-Dispersal (SND). Unconditional SND for arbitrary large H^1 data remains open.

Abstract

We record two fluids-only tools for the incompressible Navier–Stokes equations on $\mathbb{T}^3$. First, a *Ring Lemma*: if the velocity is Fourier-supported in a single Littlewood–Paley shell S_{j^*} , then on the set where vorticity is comparable to its L^2 norm the vorticity direction satisfies $\|\nabla \xi_0\|_{L^\infty(E_c)} \leq C 2^{j^*}$. Second, a *Spectral Non-Dispersal* (SND) condition requiring that a positive fraction of enstrophy remain in a dominant shell, $\inf_t J(t)/X(t) \geq c_* > 0$. Under SND we outline a conditional route to H^1 control via the Ring Lemma in the concentrated regime and enhanced dissipation in the spread regime. We do not claim unconditional global regularity.

1 Setup

Let u be a Leray–Hopf solution of

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \operatorname{div} u = 0$$

on $\mathbb{T}^3$, with $u_0 \in H^1$ divergence-free. Write Δ_j for dyadic Littlewood–Paley projections and

$$\begin{aligned} X_j(t) &= 2^{2j} \|\Delta_j u(t)\|_{L^2}^2, & X(t) &= \sum_j X_j(t) = \|\nabla u(t)\|_{L^2}^2, \\ J(t) &= \max_j X_j(t), & \rho(t) &= J(t)/X(t), & j^*(t) &= \operatorname{argmax}_j X_j(t). \end{aligned}$$

Definition 1.1 (SND). *A solution satisfies Spectral Non-Dispersal on an interval I if there exists $c_* > 0$ such that*

$$\inf_{t \in I} \frac{J(t)}{X(t)} \geq c_*.$$

Equivalently, the dominant shell always carries at least fraction c_ of the enstrophy.*

Remark 1.2. *Some companion notes use an upper-bound convention $\sup_j a_j \leq \rho_0$ for normalized shell energies a_j . Those are related but not identical; this note freezes the lower-bound form above.*

2 The Ring Lemma

Lemma 2.1 (Ring Lemma). *Suppose $u \in H^1(\mathbb{T}^3)$ has Fourier support in a single shell $S_{j^*} = \{k \in \mathbb{Z}^3 : |k| \sim 2^{j^*}\}$. Let $\omega = \text{curl } u$, $\xi_0 = \omega/|\omega|$ where $\omega \neq 0$, and*

$$E_c = \{x : |\omega(x)| \geq c\|\omega\|_{L^2}\}$$

for a fixed $c \in (0, 1)$. Then $|E_c| > 0$ for c small enough, and

$$\|\nabla \xi_0\|_{L^\infty(E_c)} \leq C 2^{j^*}$$

with C depending only on c and Bernstein constants on $\mathbb{T}^3$. The bound persists while Fourier support remains in S_{j^} .*

Proof sketch. On a single shell, Bernstein yields $\|\nabla \omega\|_{L^\infty} \lesssim 2^{j^*} \|\omega\|_{L^\infty} \lesssim 2^{2j^*} \|\omega\|_{L^2}$ (up to absolute constants). On E_c ,

$$|\nabla \xi_0| \leq \frac{|\nabla \omega|}{|\omega|} + \frac{|\nabla |\omega||}{|\omega|} \leq \frac{2\|\nabla \omega\|_{L^\infty}}{c\|\omega\|_{L^2}} \lesssim \frac{2^{j^*}}{c}.$$

Measure of E_c follows by Chebyshev. Persistence follows from persistence of Fourier support under the (smooth) dynamics on short times, or under an augmented parabolic approximation. $\square$

Corollary 2.2 (Band-limited CF control). *Under the Ring Lemma hypotheses, $\nabla \xi_0$ is bounded on E_c by a multiple of the shell frequency. In that band-limited regime, geometric vorticity-direction criteria of Constantin–Fefferman type apply with an explicit constant depending on j^* [1].*

3 Conditional regularity under SND

Theorem 3.1 (Conditional H^1 control). *Let $u_0 \in H^1(\mathbb{T}^3)$ be divergence-free and let u be an associated Leray–Hopf solution. Suppose SND holds on $[0, T]$ with constant $c_* > 0$. Then, in the concentrated regime where a dominant shell persists,*

$$u \in L^\infty([0, T]; H^1) \cap L^2([0, T]; H^2)$$

provided the Ring Lemma may be applied to a band-limited approximation of the dominant shell contribution and standard dissipative estimates control the complementary spread regime. In particular, no finite-time singularity occurs on $[0, T]$ under these hypotheses.

Proof outline. **Concentrated regime.** SND forces $J \geq c_* X$, so a dominant scale j^* carries a fixed enstrophy fraction. Approximating that contribution by a band-limited field and applying the Ring Lemma yields a vorticity-direction bound of size $O(2^{j^*})$ on E_c , feeding a CF-type geometric estimate that prevents vortex-stretching blowup while the band persists.

Spread regime. When enstrophy is dispersed across many shells, shell-spread Poincaré / enhanced dissipation estimates give additional H^1 damping (companion quantitative notes). Finite occupation time of extreme spread regimes follows from energy identities.

Glue. Alternation between regimes under SND keeps $\|\nabla u\|_{L^2}$ from blowing up on $[0, T]$. Details of constants and approximation are deferred to the longer conditional framework preprint; the present note isolates the geometric lemma and the conditional claim. $\square$

Remark 3.2 (What is not claimed). *We do not prove SND for arbitrary large data. We do not identify SND with arithmetic operators $1/\text{gcd}$ or even-sum prime-partition conditions.*

4 Open problem

Open Problem. *Prove or disprove that every Leray–Hopf solution with $u_0 \in H^1(\mathbb{T}^3)$ satisfies SND on $[0, \infty)$ for some $c_* = c_*(u_0, \nu) > 0$.*

References

- [1] P. Constantin and C. Fefferman, *Direction of vorticity and the problem of global regularity for the Navier–Stokes equations*, Indiana Univ. Math. J. **42** (1993), 775–789.
- [2] J. T. Beale, T. Kato, and A. Majda, *Remarks on the breakdown of smooth solutions for the 3-D Euler equations*, Comm. Math. Phys. **94** (1984), 61–66.